

Advanced Calculus

Differential Equations (DE):

An equation involving derivatives of one or more dependent variables with respect to one or more independent variables is called a differential equation.

Ordinary Differential Equations (ODE):

A differential equation involving ordinary derivatives of one or more dependent variables with respect to a single independent variable is called an ordinary differential equation.

Partial Differential Equation (PDE):

A differential equation involving partial derivatives of one or more dependent variables with respect to more than one independent variable is called a partial differential equation.

EXAMPLES BY CATEGORY

Ordinary Differential Equations (ODE)

$$\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx} \right)^2 = 0$$

$$\frac{d^4x}{dt^4} + 5\frac{d^2x}{dt^2} + 3x = \sin(t)$$

Partial Differential Equations (PDE)

$$\frac{\partial v}{\partial \sigma} + \frac{\partial v}{\partial t} = v$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

Not DE but Basic Derivatives

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

Order of a DE:

The order of the highest-order derivative involved in a differential equation is called the order of the differential equation.

Degree of a DE:

The degree of a differential equation is the power of the highest-order derivative present in the equation.

EXAMPLES BY CATEGORY

1st Order 1st Degree ODE

$$\frac{dy}{dx} + y = e^x$$

2nd Order 1st Degree ODE

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$$

2nd Order 2nd Degree ODE

$$\left(\frac{d^2y}{dx^2}\right)^2 + y\frac{dy}{dx} = 0$$

1st Order 1st Degree PDE

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0$$

2nd Order 1st Degree PDE

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$$

2nd Order 2nd Degree PDE

$$\left(\frac{\partial^2 u}{\partial x^2}\right)^2 + \left(\frac{\partial^2 u}{\partial y^2}\right)^2 = 0$$

Linear ODE:

A linear ordinary differential equation of order n , in the dependent variable y and the independent variable x , is an equation that can be expressed in the form.

$$a_0 y^{(n)} + a_1 y^{(n-1)} + \dots + a_n y = f(x)$$

or equivalently,

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n(x) y = f(x)$$

where $a_0, a_1, \dots, a_n$ are functions of x (not identically zero) and $f(x)$ is a given function.

Important Observations:

- 1) The dependent variable y and its various derivatives occur to the first degree only.
- 2) No products of y and/or any of its derivatives are present.
- 3) No transcendental functions of y or its derivatives occur (e.g., $\sin y$, e^y).

Nonlinear ODE:

A nonlinear ordinary differential equation is an ordinary differential equation that is not linear.

EXAMPLES BY CATEGORY

Linear ODE with Constant Coefficients

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$$

Linear ODE with Variable Coefficients

$$\frac{d^4y}{dx^4} + x^{30}\frac{dy}{dx} + 2y = xe^x$$

Nonlinear ODE Examples

$$\left(\frac{d^2y}{dx^2}\right)^2 + y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} + \sin(y) = 0$$

$$A\frac{d^2y}{dx^2} + K\frac{d^3y}{dx^3} + x\frac{dy}{dx} = xe^x\frac{dx}{dy}$$

General Form of a Nonlinear ODE

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right) = 0$$

What Makes an ODE Nonlinear?

An ODE is classified as nonlinear if it violates any of the key linear criteria:

- **Higher Powers:** Any y or derivative term has an exponent other than 1.
- **Product Terms:** The variable y is multiplied by its own derivatives.
- **Nonlinear Functions:** The equation contains functions like $\sin(y)$, e^y , or $\ln(y)$.

Origin and Application:

Differential equations occur in connection with numerous problems encountered across various branches of science and engineering:

- Population Dynamics
- Dynamical Systems and Chaos
- Fluid Dynamics
- Heterogeneous Heat Transfer
- Biophysics and Mathematical Biology

Formation of Differential Equations:

To form a differential equation from a given relation involving independent/dependent variables and arbitrary constants, we eliminate the arbitrary constants by successive differentiation.

Example Problem:

Form the differential equation by eliminating the constants a and b from:

$$y = ax + bx^2 \quad (1)$$

Solution:

Given the relation with two arbitrary constants a and b :

$$y = ax + bx^2 \quad (2)$$

Since there are two arbitrary constants, we differentiate the equation twice with respect to x to eliminate them.

Step 1: First Differentiation Differentiating Equation (2) with respect to x :

$$\frac{dy}{dx} = a + 2bx \quad (3)$$

Step 2: Second Differentiation Differentiating Equation (3) with respect to x :

$$\frac{d^2y}{dx^2} = 2b \implies b = \frac{1}{2} \frac{d^2y}{dx^2} \quad (4)$$

Step 3: Eliminate Constant a Substitute the value of b from Equation (4) into Equation (3):

$$\begin{aligned} \frac{dy}{dx} &= a + 2 \left(\frac{1}{2} \frac{d^2y}{dx^2} \right) x \\ \frac{dy}{dx} &= a + x \frac{d^2y}{dx^2} \implies a = \frac{dy}{dx} - x \frac{d^2y}{dx^2} \end{aligned}$$

Step 4: Substitute a and b into the Original Equation Substitute the expressions for a and b back into Equation (2):

$$y = \left(\frac{dy}{dx} - x \frac{d^2y}{dx^2} \right) x + \left(\frac{1}{2} \frac{d^2y}{dx^2} \right) x^2$$

Expand the terms:

$$y = x \frac{dy}{dx} - x^2 \frac{d^2y}{dx^2} + \frac{1}{2} x^2 \frac{d^2y}{dx^2}$$

Combine the like terms of $\frac{d^2y}{dx^2}$:

$$y = x \frac{dy}{dx} - \frac{1}{2} x^2 \frac{d^2y}{dx^2}$$

Final Form: Multiply the entire equation by 2 and rearrange terms to obtain the standard form:

$$x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0 \quad (5)$$

This is the required ordinary differential equation of 2^{nd} order and 1^{st} degree.

1 Equations Reducible to Homogeneous Form

A differential equation of the form

$$\frac{dy}{dx} = \frac{ax + by + c}{a_1x + b_1y + c_1}, \quad \text{where } \frac{a}{a_1} \neq \frac{b}{b_1}$$

can be reduced to a homogeneous form by putting $x = X + h$ and $y = Y + k$. Then, $\frac{dY}{dX} = \frac{dy}{dx}$.

EXAMPLE: REDUCIBLE TO HOMOGENEOUS FORM

Solve: $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$ (1)

Solution:Put $x = X + h$ and $y = Y + k$, where h and k are constants.Then, $\frac{dy}{dx} = \frac{dY}{dX}$.

Substituting into (1):

$$\frac{dY}{dX} = \frac{X+h+2(Y+k)-3}{2(X+h)+(Y+k)-3} = \frac{X+2Y+(h+2k-3)}{2X+Y+(2h+k-3)}$$

Now, choose h and k so that:

$$h+2k-3=0 \quad \text{and} \quad 2h+k-3=0$$

Solving these equations, we get $h=1$ and $k=1$.

Therefore, the equation becomes:

$$\frac{dY}{dX} = \frac{X+2Y}{2X+Y}$$

which is homogeneous in X and Y .Put $Y = vX$, so that $\frac{dY}{dX} = v + X \frac{dv}{dX}$.

$$v + X \frac{dv}{dX} = \frac{X+2vX}{2X+vX} = \frac{1+2v}{2+v}$$

$$X \frac{dv}{dX} = \frac{1+2v}{2+v} - v = \frac{1+2v-2v-v^2}{2+v} = \frac{1-v^2}{2+v}$$

Separating the variables:

$$\frac{dX}{X} = \left(\frac{2+v}{1-v^2} \right) dv = \left(\frac{2}{1-v^2} + \frac{v}{1-v^2} \right) dv$$

Integrating both sides:

$$\begin{aligned} \ln X &= 2 \cdot \frac{1}{2} \ln \left(\frac{1+v}{1-v} \right) - \frac{1}{2} \ln(1-v^2) + \ln c \\ \Rightarrow X &= c \left(\frac{1+v}{1-v} \cdot \frac{1}{\sqrt{1-v^2}} \right) = c \left(\frac{1+v}{1-v} \cdot \frac{1}{\sqrt{1+v}\sqrt{1-v}} \right) \\ &\Rightarrow X = \frac{c\sqrt{1+v}}{(1-v)^{3/2}} \end{aligned}$$

Squaring both sides:

$$X^2 = \frac{c^2(1+v)}{(1-v)^3} \implies X^2(1-v)^3 = c^2(1+v)$$

Substituting $v = \frac{Y}{X}$:

$$\begin{aligned} X^2 \left(1 - \frac{Y}{X} \right)^3 &= c^2 \left(1 + \frac{Y}{X} \right) \\ \Rightarrow X^2 \frac{(X-Y)^3}{X^3} &= c^2 \left(\frac{X+Y}{X} \right) \implies (X-Y)^3 = c^2(X+Y) \end{aligned}$$

Finally, substituting back $X = x-1$ and $Y = y-1$:

$$\{(x-1)-(y-1)\}^3 = c^2\{(x-1)+(y-1)\}$$

$$\Rightarrow (x-y)^3 = c^2(x+y-2)$$

2 Linear Differential Equations

A differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

is called a linear differential equation of the first order.

Method of Solution:

To solve the equation $\frac{dy}{dx} + P(x)y = Q(x)$ (1), we multiply both sides by an **Integrating Factor (I.F.)**.

The Integrating Factor is given by $e^{\int P(x)dx}$.

Multiplying (1) by the I.F.:

$$e^{\int P(x)dx} \frac{dy}{dx} + P(x)y e^{\int P(x)dx} = Q(x) e^{\int P(x)dx}$$

Notice that the left-hand side is the exact derivative of the product $y \cdot e^{\int P(x)dx}$:

$$\frac{d}{dx} \left(y e^{\int P(x)dx} \right) = Q(x) e^{\int P(x)dx}$$

Integrating with respect to x :

$$y e^{\int P(x)dx} = \int Q(x) e^{\int P(x)dx} dx + c$$

which is the required general solution.

EXAMPLE: LINEAR DE

Solve: $x \frac{dy}{dx} + 2y = x^2 \log x$ (1)

Solution:

Dividing by x , we put the equation in standard linear form:

$$\frac{dy}{dx} + \frac{2}{x}y = x \log x$$

Here, $P(x) = \frac{2}{x}$ and $Q(x) = x \log x$.

The Integrating Factor (I.F.) is:

$$\text{I.F.} = e^{\int \frac{2}{x} dx} = e^{2 \log x} = e^{\log x^2} = x^2$$

Multiplying both sides of the standard equation by x^2 :

$$x^2 \frac{dy}{dx} + 2xy = x^3 \log x$$

$$\Rightarrow \frac{d}{dx}(yx^2) = x^3 \log x$$

Integrating both sides:

$$x^2 y = \int x^3 \log x \, dx + c$$

Using integration by parts (taking $\log x$ as the first function and x^3 as the second):

$$x^2 y = \log x \cdot \frac{x^4}{4} - \int \frac{1}{x} \cdot \frac{x^4}{4} dx + c$$

$$\Rightarrow x^2 y = \frac{x^4}{4} \log x - \frac{1}{4} \int x^3 dx + c$$

$$\Rightarrow x^2 y = \frac{x^4}{4} \log x - \frac{1}{4} \cdot \frac{x^4}{4} + c = \frac{x^4}{4} \left(\log x - \frac{1}{4} \right) + c$$

Dividing by x^2 :

$$y = cx^{-2} + \frac{1}{4}x^2 \left(\log x - \frac{1}{4} \right)$$

which is the required solution.

EXAMPLE: INITIAL VALUE PROBLEM (IVP)

Solve the IVP: $(x^2 + 1)\frac{dy}{dx} + 4xy = x, \quad y(2) = 1$ (2)

Solution:

Dividing by $(x^2 + 1)$, we obtain the standard linear form:

$$\frac{dy}{dx} + \frac{4x}{x^2 + 1}y = \frac{x}{x^2 + 1} \quad (3)$$

Here, $P(x) = \frac{4x}{x^2 + 1}$. The Integrating Factor is:

$$\text{I.F.} = e^{\int \frac{4x}{x^2 + 1} dx} = e^{2 \ln(x^2 + 1)} = e^{\ln((x^2 + 1)^2)} = (x^2 + 1)^2$$

Multiplying equation (3) by the I.F.:

$$\begin{aligned} (x^2 + 1)^2 \frac{dy}{dx} + \frac{4x}{x^2 + 1} (x^2 + 1)^2 y &= \frac{x}{x^2 + 1} (x^2 + 1)^2 \\ \Rightarrow (x^2 + 1)^2 \frac{dy}{dx} + 4x(x^2 + 1)y &= x(x^2 + 1) \\ \Rightarrow \frac{d}{dx} ((x^2 + 1)^2 y) &= x^3 + x \end{aligned}$$

Integrating both sides:

$$\begin{aligned} (x^2 + 1)^2 y &= \int (x^3 + x) dx + c \\ \Rightarrow (x^2 + 1)^2 y &= \frac{x^4}{4} + \frac{x^2}{2} + c \quad (4) \end{aligned}$$

Applying the initial condition $y(2) = 1$ (i.e., when $x = 2, y = 1$):

$$\begin{aligned} (2^2 + 1)^2 \cdot 1 &= \frac{2^4}{4} + \frac{2^2}{2} + c \implies 5^2 = \frac{16}{4} + \frac{4}{2} + c \\ 25 &= 4 + 2 + c \implies c = 19 \end{aligned}$$

Substituting $c = 19$ back into equation (4):

$$(x^2 + 1)^2 y = \frac{x^4}{4} + \frac{x^2}{2} + 19$$

This is the required solution.

3 Bernoulli Equations

A differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad (1)$$

is called a Bernoulli differential equation.

Method of Solution:

Divide equation (1) by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{-n+1} = Q(x) \quad (2)$$

Put $y^{-n+1} = v$. Then differentiating with respect to x :

$$(-n+1)y^{-n}\frac{dy}{dx} = \frac{dv}{dx} \implies y^{-n}\frac{dy}{dx} = \frac{1}{1-n}\frac{dv}{dx}$$

Substitute this into equation (2):

$$\begin{aligned}\frac{1}{1-n}\frac{dv}{dx} + P(x)v &= Q(x) \\ \implies \frac{dv}{dx} + (1-n)P(x)v &= (1-n)Q(x)\end{aligned}$$

This is a linear differential equation in v and x , which can be solved using the integrating factor method.

General Case:

An equation of the form $f'(y)\frac{dy}{dx} + P(x)f(y) = Q(x)$ can be reduced to a linear form by putting $f(y) = v$, which gives $f'(y)\frac{dy}{dx} = \frac{dv}{dx}$. The equation becomes:

$$\frac{dv}{dx} + P(x)v = Q(x)$$

which is linear in v and x .

EXAMPLE: BERNOULLI EQUATION

Solve: $\frac{dy}{dx} + y = xy^3$ (1)

Solution:

This is a Bernoulli differential equation with $n = 3$. Dividing by y^3 :

$$y^{-3} \frac{dy}{dx} + y^{-2} = x$$

Let $v = y^{1-n} = y^{1-3} = y^{-2}$. Then differentiating with respect to x :

$$\frac{dv}{dx} = -2y^{-3} \frac{dy}{dx} \implies y^{-3} \frac{dy}{dx} = -\frac{1}{2} \frac{dv}{dx}$$

Substitute this back into the equation:

$$-\frac{1}{2} \frac{dv}{dx} + v = x \implies \frac{dv}{dx} - 2v = -2x \quad (2)$$

This is a linear differential equation in v and x . Here, $P(x) = -2$.

The Integrating Factor is:

$$\text{I.F.} = e^{\int -2dx} = e^{-2x}$$

Multiplying equation (2) by the I.F.:

$$\begin{aligned} e^{-2x} \frac{dv}{dx} - 2ve^{-2x} &= -2xe^{-2x} \\ \implies \frac{d}{dx} (ve^{-2x}) &= -2xe^{-2x} \end{aligned}$$

Integrating both sides:

$$ve^{-2x} = \int -2xe^{-2x} dx + c$$

Using integration by parts:

$$\begin{aligned} ve^{-2x} &= -2 \left[x \left(\frac{e^{-2x}}{-2} \right) - \int 1 \cdot \left(\frac{e^{-2x}}{-2} \right) dx \right] + c \\ ve^{-2x} &= xe^{-2x} - \int e^{-2x} dx + c = xe^{-2x} + \frac{1}{2}e^{-2x} + c \end{aligned}$$

Dividing by e^{-2x} :

$$v = x + \frac{1}{2} + ce^{2x}$$

where c is an arbitrary constant.

Since $v = y^{-2} = \frac{1}{y^2}$, we obtain the solution in the form:

$$\frac{1}{y^2} = x + \frac{1}{2} + ce^{2x}$$

EXAMPLE: BERNOULLI EQUATION

Solve: $x \frac{dy}{dx} + y = y^2 \log x$ (1)

Solution:

Dividing by xy^2 :

$$\frac{1}{x} y^{-2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = \frac{1}{x} \log x$$

Actually, dividing the whole equation by y^2 first and then dividing the whole equation by x :

$$y^{-2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = \frac{1}{x} \log x$$

Put $-y^{-1} = v \implies y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$.

Substitute into the equation:

$$\frac{dv}{dx} - \frac{1}{x} v = \frac{1}{x} \log x \quad (2)$$

The Integrating Factor is:

$$\text{I.F.} = e^{\int -\frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$$

Multiplying equation (2) by the I.F.:

$$\begin{aligned} \frac{1}{x} \frac{dv}{dx} - \frac{1}{x^2} v &= \frac{1}{x^2} \log x \\ \implies \frac{d}{dx} \left(v \cdot \frac{1}{x} \right) &= \frac{1}{x^2} \log x \end{aligned}$$

Integrating both sides:

$$v \cdot \frac{1}{x} = \int \frac{1}{x^2} \log x \, dx + c$$

Using integration by parts:

$$\begin{aligned} \int x^{-2} \log x \, dx &= \log x \left(\frac{x^{-1}}{-1} \right) - \int \frac{1}{x} \left(\frac{x^{-1}}{-1} \right) dx \\ &= -\frac{1}{x} \log x + \int x^{-2} dx = -\frac{1}{x} \log x - \frac{1}{x} \end{aligned}$$

So we have:

$$\frac{v}{x} = c - \frac{1}{x} \log x - \frac{1}{x} = c - \frac{1 + \log x}{x}$$

Multiplying by x :

$$v = cx - (1 + \log x)$$

Since $v = -y^{-1} = -\frac{1}{y}$:

$$-\frac{1}{y} = cx - (1 + \log x) \implies \frac{1}{y} = (1 + \log x) - cx$$

This is the required solution.

4 Exact Differential Equations

Total Differential:

If the function $F(x, y)$ has continuous first partial derivatives in a domain D , then the total differential dF is defined by:

$$dF(x, y) = \frac{\partial F(x, y)}{\partial x} dx + \frac{\partial F(x, y)}{\partial y} dy \quad \forall (x, y) \in D$$

Exact Differential:

The expression $M(x, y)dx + N(x, y)dy$ is called an exact differential if there exists a function $F(x, y)$ such that:

$$dF(x, y) = M(x, y)dx + N(x, y)dy$$

Theorem (Necessary Condition):

Consider the DE $M(x, y)dx + N(x, y)dy = 0$ where $M(x, y)$ and $N(x, y)$ have continuous 1st partial derivatives $\forall (x, y) \in D$.

If the DE (1) is exact in D , then $\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x} \quad \forall (x, y) \in D$.

Conversely, if $\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x} \quad \forall (x, y) \in D$, then the DE (1) is exact in D .

Working Rule:

If the DE $M(x, y)dx + N(x, y)dy = 0$ satisfies the condition $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then it is exact.

To integrate it:

1. Integrate M with respect to x regarding y as a constant.
2. Find out those terms in N which are free from x and integrate them with respect to y .
3. Add the two expressions so obtained and equate the sum to an arbitrary constant.

This gives the general solution of the given exact DE.

EXAMPLE: EXACT DE

Solve: $(3x^2 + 4xy)dx + (2x^2 + 2y)dy = 0$

Solution:

Here, $M(x, y) = 3x^2 + 4xy$ and $N(x, y) = 2x^2 + 2y$.

$$\frac{\partial M}{\partial y} = 4x \quad \text{and} \quad \frac{\partial N}{\partial x} = 4x$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the given equation is exact in every rectangular domain D .
According to the working rule:

1. Integrate M with respect to x regarding y as a constant:

$$\int M(x, y)dx = \int (3x^2 + 4xy)dx = x^3 + 2x^2y$$

2. The terms in $N(x, y)$ free from x are $2y$. Integrate this with respect to y :

$$\int 2y \, dy = y^2$$

Adding these two and equating to a constant:

$$x^3 + 2x^2y + y^2 = c$$

which is the required solution.

Note: You may solve it as it's given in J.L. Ross.

EXAMPLE: EXACT DE (IVP)

Solve the IVP: $(2x \cos y + 3x^2 y)dx + (x^3 - x^2 \sin y - y)dy = 0$, $y(0) = 2$

Solution:

$$\frac{\partial M}{\partial y} = -2x \sin y + 3x^2 = \frac{\partial N}{\partial x}$$

The given DE is exact.

① Integrate M with respect to x treating y as constant:

$$\int (2x \cos y + 3x^2 y) dx = x^2 \cos y + x^3 y$$

② Integrate terms in N free from x :

$$\int -y dy = -\frac{y^2}{2}$$

Adding these two and equating to a constant c :

$$x^2 \cos y + x^3 y - \frac{y^2}{2} = c$$

At $x = 0, y = 2$:

$$0 \cdot \cos 2 + 0 \cdot 2 - \frac{2^2}{2} = c \implies c = -2$$

Therefore, the solution is:

$$x^2 \cos y + x^3 y - \frac{y^2}{2} = -2$$

5 Integrating Factor (Non-Exact DE)

If the DE $M(x, y)dx + N(x, y)dy = 0$ (1)
is not exact, but it becomes exact when multiplied by $\mu(x, y)$:

$$\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0 \quad (2)$$

then $\mu(x, y)$ is called an **integrating factor (I.F.)** of (1).

Rules for finding I.F.:

R.1) If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$, a function of x only, then $e^{\int f(x)dx}$ is an I.F.

R.2) If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M} = g(y)$, a function of y only, then $e^{\int g(y)dy}$ is an I.F.

R.3) If $Mdx + Ndy = 0$ is homogeneous and $Mx + Ny \neq 0$, then $\frac{1}{Mx+Ny}$ is an I.F.

EXAMPLE: USING INTEGRATING FACTOR

Solve: $(x^2 + y^2 + 1)dx - 2xydy = 0$ (1)

Solution:

$$M = x^2 + y^2 + 1, \quad N = -2xy;$$

$$\frac{\partial M}{\partial y} = 2y, \quad \frac{\partial N}{\partial x} = -2y$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \implies \text{DE (1) is not exact.}$$

However,

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y - (-2y)}{-2xy} = \frac{-2}{x}$$

which is a function of x alone.

$$\text{I.F.} = e^{\int \frac{-2}{x} dx} = e^{-2 \log x} = \frac{1}{x^2}$$

Multiplying by $\frac{1}{x^2}$, DE (1) becomes:

$$\left(1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right) dx - \frac{2y}{x^2} dy = 0 \quad (2)$$

$$\text{Now, } \frac{\partial M}{\partial y} = \frac{2y}{x^2} \text{ \& } \frac{\partial N}{\partial x} = \frac{2y}{x^2} \implies \text{DE is exact.}$$

Now, Integrating $\left(1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right)$ w.r.t x , keeping y as constant, we get:

$$x - \frac{y^2}{x} - \frac{1}{x}$$

and in $\frac{-2y}{x^2}$ there is no term free from x .

Hence the solution is:

$$x - \frac{y^2}{x} - \frac{1}{x} = c$$

Ans.

EXAMPLE: USING INTEGRATING FACTOR

Solve: $(x^2 + y^2 + 2x)dx + 2ydy = 0$ (1)

Solution:

$$\frac{\partial M}{\partial y} = 2y \text{ \& } \frac{\partial N}{\partial x} = 0$$

$$\text{Now, } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y - 0}{2y} = 1$$

$$\text{I.F.} = e^{\int 1 dx} = e^x$$

$$\text{Multiplying (1) by } e^x \implies e^x(x^2 + y^2 + 2x)dx + e^x 2ydy = 0$$

which is exact.

This equation can be written as:

$$e^x(x^2 + 2x)dx + (y^2 e^x)dx + e^x 2ydy = 0$$

$$\implies d(x^2 e^x) + d(y^2 e^x) = 0$$

Integrating, we get:

$$x^2 e^x + y^2 e^x = c$$

Ans.

6 Trajectories

Let $F(x, y, c) = 0$ (1)

be a given one-parameter family of curves.

A curve that intersects the curves of the family (1) at a constant angle $\alpha \neq 90^\circ$ is called an **oblique trajectory** (or α trajectory) of the given family.

A curve that intersects the curves of the family at right angles ($\alpha = 90^\circ$) is called an **orthogonal trajectory** of the given family.

Equations of the trajectory: If a family of curves be given by the DE $f(x, y, p) = 0$ then, its α trajectory is given by:

$$f\left(x, y, \frac{p - \tan \alpha}{1 + p \tan \alpha}\right) = 0$$

Orthogonal trajectories: If $\alpha = 90^\circ$, then

$$\frac{p - \tan \alpha}{1 + p \tan \alpha} = \frac{p \cot \alpha - 1}{\cot \alpha + p} = \frac{-1}{p} \quad [\cot 90^\circ = 0]$$

Hence, corresponding to the family of curves whose DE is $f(x, y, p) = 0$, the differential equation of the orthogonal trajectory is $f\left(x, y, -\frac{1}{p}\right) = 0$, which on integration, gives a family of trajectories orthogonal to the given family of curves.

Polar coordinates: If a family of curves be given by $f(r, \theta, \frac{dr}{d\theta}) = 0$ then the orthogonal trajectory is given by $f\left(r, \theta, -r^2 \frac{d\theta}{dr}\right) = 0$.

EXAMPLE: ORTHOGONAL TRAJECTORIES

Find the orthogonal trajectories of the family of circles $x^2 + y^2 = c^2$ (1)

Solution:

Family of circles is $x^2 + y^2 = c^2$ (1)

Differentiating $\Rightarrow 2x + 2y \frac{dy}{dx} = 0$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{y} \quad (2)$$

which is the DE of the given family (1).

We now replace $\frac{dy}{dx}$ by its negative reciprocal $-\frac{dx}{dy}$ in the DE (2) to obtain the differential equation of the orthogonal trajectories:

$$\frac{dy}{dx} = \frac{y}{x} \quad (3)$$

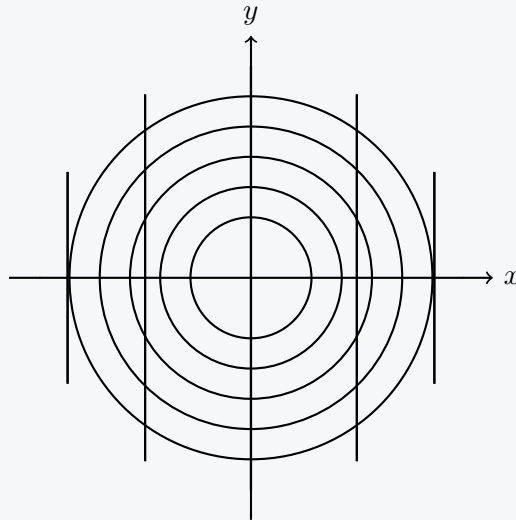
(the orthogonal trajectories DE).

We now solve (3). Separating variables, we have:

$$\frac{dy}{y} = \frac{dx}{x}$$

Integrating $\Rightarrow \ln y = \ln x + \ln k \Rightarrow y = kx$.

This is one parameter family of solutions of DE (3) and thus represents the family of orthogonal trajectories of the given family of circles (1).



EXAMPLE: ORTHOGONAL TRAJECTORIES**Find the orthogonal trajectories of $y = cx^2$** **Solution:**Differentiating $\Rightarrow \frac{dy}{dx} = 2cx = 2\left(\frac{y}{x^2}\right) \cdot x = \frac{2y}{x}$ Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ we get,

$$-\frac{dx}{dy} = \frac{2y}{x}$$

which is the DE of orthogonal trajectory $\Rightarrow 2ydy + xdx = 0$.Integrating, $y^2 + \frac{x^2}{2} = k_0 \Rightarrow$ (Separating the vars)

$$\Rightarrow 2y^2 + x^2 = 2k_0 = k^2$$

which is the family of orthogonal trajectories of the given family $y = cx^2$.**EXAMPLE: OBLIQUE TRAJECTORIES****Find a family of oblique trajectory of the family of straight lines $y = cx$ at angle $\alpha = 45^\circ$. (S.L. Ross p75)**

7 Higher Order Linear DE

A LODE (Linear Ordinary Differential Equation) of order n in the dependent variable y and the independent variable x is an equation that can be expressed in the form,

$$a_0(x)\frac{d^ny}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_{n-1}(x)\frac{dy}{dx} + a_n(x)y = F(x) \quad (6)$$

where $a_0(x) \neq 0$.

We shall assume that $a_0, a_1, \dots, a_n$ are continuous on an interval $a \leq x \leq b$ & $a_0 \neq 0 \forall x$ on $a \leq x \leq b$.

The right-hand member $F(x)$ is called the **non-homogeneous term**.If $F(x) = 0$ then eqn (1) reduces to:

$$a_0(x)\frac{d^ny}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_{n-1}(x)\frac{dy}{dx} + a_n(x)y = 0 \quad (7)$$

and is then called **homogeneous**.

Let x_0 be any point on $a \leq x \leq b$ and let $c_0, c_1, \dots, c_{n-1}$ be arbitrary constants of n . There exists a unique solution f^* of Eqn (1) s.t.

$$f(x_0) = c_0, f'(x_0) = c_1, \dots, f^{(n-1)}(x_0) = c_{n-1}$$

which is defined on $a \leq x \leq b$.**Note:** Let f be a solution of (2) then $f(x_0) = 0, f'(x_0) = 0, \dots, f^{(n-1)}(x_0) = 0$.